

Sustainability Aware Asset Management

Portfolio Allocation with a Carbon Objective

May 29, 2026

Lukas TONKOVIC

lukas.tonkovic@unil.ch

22-608-855

Marko ZIVALJEVIC

marko.zivaljevic@unil.ch

22-436-034

Philippe BORGHINI VILLIGER

philippe.borghinivilliger@unil.ch

21-420-229

Stefan STEVANOVIC

stefan.stevanovic@unil.ch

22-504-591



UNIL | Université de Lausanne

HEC Lausanne

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1 Introduction

Climate change has in recent decades become one of the most talked about phenomena in the world. NASA defines it as “a long-term change in the average weather patterns that have come to define Earth’s local, regional and global climates.” Climate change is not new as it has naturally existed since the dawn of time due to phenomena like volcanic eruptions or variations in the solar cycle. However, since the 1800s, climate change has started accelerating at a much more alarming rate, with human activity being considered the main cause behind this acceleration. The Industrial Revolution, although representing a beam of technological hope and economic growth, also marked the beginning of a sharp increase in greenhouse gas emissions, ultimately contributing to global warming and environmental degradation.

As climate change started gaining more and more mainstream attention, governments and international institutions progressively began implementing long-term strategies aimed at limiting its effects. The Kyoto Protocol, adopted in 1997, was one of the first major international agreements attempting to regulate greenhouse gas emissions in industrialized countries. In the following years, scientists increasingly began to understand the economic, environmental and social consequences global warming would have. To that extent, the 2°C limit quickly became the threshold not to exceed. This objective was later officially integrated into the Paris Agreement, which aims to keep “the increase in the global average temperature to well below 2°C above pre-industrial levels” over the long term.

Economic and financial agents slowly became more involved in the matter as firms increasingly had to abide by environmental regulations, often leading to higher production and supply chain costs through the use of more environmentally friendly materials and technologies. This ultimately ended up having a downstream effect on investors through stricter regulation, environmental taxes, and transition costs affecting firms’ profitability. Additionally, it is now becoming more important than ever for managers and investors to know where their factories are located and where their resources are coming from, as extreme weather phenomena become increasingly frequent with each passing year. That being said, investors have progressively begun considering environmental implications when constructing their portfolios. Today, many institutional investors pay close attention to the greenhouse gas emissions and carbon exposure of the companies they may invest in.

Emerging countries, being characterized by rapid growth and strong industrialization, have always been a major hub for national and international investment. Investors see these countries as gold mines where substantial returns can potentially be made. Nonetheless, as attractive as these economies are, they are also among the least prepared to face global warming. Many of these countries, being located in geographically vulnerable areas around the world, tend to be more exposed to extreme weather phenomena accelerated by global warming. Furthermore, their infrastructure tends to be weaker and less suited to handle such catastrophes. Financial buffers are scarce, forcing countries to rely heavily on international financial aid following disasters, e.g., the Turkey-Syria earthquake in 2023. Consequently, emerging economies become an even greater focal point for investors when considering climate-related risks and sustainable investment strategies.

Investors have narrowed their focus to greenhouse gas emissions when focusing on environmentally friendly strategies, as these have been outlined to be the main contributors of climate change. Defined as “a category of gases that absorb heat energy emitted from the planet’s surface and remain in the Earth’s atmosphere for a long time”, greenhouse gases can be separated into 3 scopes. Scope 1 GHG are the direct emissions coming from the operations and activities of a company. Scope 2 GHG, on the other hand, are indirect emissions coming from the consumption of purchased energy. Many investors have implemented strategies aiming at

reducing the emissions of these two scopes over the years, like the purchase of green financial instruments, capital reallocation, etc. Nonetheless, despite these strategies trending in recent years, many investors still come across several challenges when attempting to effectively reduce portfolio emissions.

We will therefore aim to implement climate-aware asset management concepts into investors' portfolio strategies. We will particularly focus on Scope 1 and Scope 2 greenhouse gas emissions in emerging countries, given how economically important these economies are globally and how vulnerable they can be to climate change. Different portfolio strategies will be tested in order to assess how effectively they reduce greenhouse gas emissions and how they affect portfolio performance financially, as well as the limitations they may present.

2 Literature Review

Climate change has been at the forefront of most domains in recent years. The question, however, is whether it also affects investors and other economic actors. To what extent should the average investor actually care about climate change when constructing a portfolio? As per *The Importance of Climate Risks for Institutional Investors* [1], investors do in fact care about climate change as they believe stock prices do not fully account for climate risks such as carbon taxes, environmental regulation, and future transition costs. This can make highly polluting firms appear more valuable than they actually are, potentially leading to valuation cuts in the future. Consequently, investors actively monitor the greenhouse gas emissions of their portfolios and progressively attempt to reduce their exposure to highly polluting firms while still maintaining portfolio diversification, in order to avoid these additional and often unpredictable costs. However, when investors do invest in polluting firms, they carefully take these risks into consideration. As stated in *Do Investors Care about Climate Risk?* [2], "Our evidence is that investors are discerning these cross-sectional differences and are pricing in carbon risk." Finally, it is important to note that investors can be affected by climate change through channels other than asset valuation alone. For example, physical risks such as extreme weather events can directly affect the supply and demand of a company's products or services. More generally, *Climate change: the impact for investors* [3] explains that "Climate change creates a spectrum of risks, often classified as adaptation, physical and transition risks. These create a myriad of potential financial risks for investors and companies."

We can now ask ourselves whether investors actually do play a major role in climate change on the global stage. *The Big Book of Climate Investing* [4] states that "Investors can play a major role in directing how money is channeled towards those companies making a difference to climate change", meaning that investors can significantly influence the problem through their financing decisions. By choosing which firms to finance and which ones not to finance, investors can indirectly encourage more sustainable corporate behaviour while penalizing highly polluting activities. Furthermore, *Four Ways Investors Can Act on Climate Change* [5] explains that "Morgan Stanley Financial Advisors can help identify opportunities for climate solutions investments, such as companies that are developing new and innovative technologies for renewable and alternative energy sources." This highlights the fact that investors do not only reduce exposure to polluting firms, but can also actively support the development of sustainable technologies and low-carbon solutions. Finally, it is important to focus on the scale of financing required for the climate transition. In *Private Equity and Public Markets: an Investors' Guide to Climate Impact Investing* [6], the authors stress that "To fund a transition able to keep the planet below a temperature rise of 1.5 degrees Celsius, the world must increase spending on solutions to at least \$4.13 trillion every year by 2030." Consequently, investors are expected to play a central role in financing sustainable technologies and climate-transition initiatives over the coming decades.

We can now focus on emerging countries and analyze how severely they are affected by climate change compared to more developed economies. In *Climate change and development in three charts* [7], the authors explain that developing countries are expected to be hit harder by climate change and extreme weather events, stating that “all the evidence points to developing countries being hardest hit by the impacts of climate change: they are already hotter, more exposed to extreme weather events, more vulnerable to adverse impacts on health and agriculture, less resilient due to weaker infrastructure and institutions, and face larger reductions in future growth and incomes.” This is corroborated by *Emerging Economies and Climate Change* [8], which states that “Despite being less responsible for climate change, emerging countries are more vulnerable to its effects than their advanced counterparts.”

Scope 1 and Scope 2 emissions capture a firm’s operational carbon footprint, in contrast to Scope 3 emissions, which capture emissions generated throughout the value chain. We can therefore ask ourselves how much Scope 1 and 2 emissions actually represent on the global scale of greenhouse gas emissions. In *What are Scope 1, 2 and 3 emissions?* [9], McKinsey estimates that “Scope 3 emissions typically represent around 90 percent of a company’s total emissions, though the figure can vary by industry and company.” This suggests that Scope 1 and Scope 2 emissions together only account for a relatively small fraction of total greenhouse gas emissions.

This leads us to the existing strategies used to reduce Scope 1 and Scope 2 emissions in emerging countries. One of the main approaches is asset reallocation. Investors reduce investments in firms with high operational emissions (Scope 1 + 2) and increase their exposure to firms with lower carbon intensity, as highlighted in *Building benchmark portfolios with decreasing carbon footprints* [10].

Nonetheless, we must also examine the limitations that these strategies may have. *Performance and Challenges of Net-Zero Strategies in the Context of the EU Regulation* [11] stresses that as portfolios become more decarbonized, their performance may substantially deviate from the benchmark portfolio they are trying to follow, therefore contributing to larger tracking errors.

It’s important to raise questions on how the portfolio’s performance can be affected by green investing. *ESG Investing and Climate Transition* [12] has shown that ESG approaches don’t consistently support better financial outcomes.

Finally one can wonder how asset allocation changes when adopting an environmentally friendly investment strategy. *ESG/Green Investment and Allocation of Portfolio Assets* [13] argues that asset allocation under ESG can differ drastically from traditional portfolio allocation because firms are now evaluated based on environmental and sustainability-related metrics rather than purely financial metrics.

3 Data and Methodology

3.1 Data and Preprocessing

The data on financial returns and carbon, used in this project, cover the period from 1999 to 2025 for financial metrics and from 2002 to 2025 for information on carbon emissions. Namely, firms’ revenues, market capitalization, and total return indices (RI), also known as price indices, fall under financial metrics. Scope 1 and 2 emissions evidently come under carbon data. However, the limited nature of information on carbon emissions before 2010 means portfolio construction begins from 2014 onwards. Firstly, the complete list of companies is analyzed and categorized

in such a manner that only the enterprises in the region of interest, namely emerging markets, remain. Therefore, our work is based on a sample consisting of 702 firms operating in emerging markets, as given after categorizing the data on all companies. Furthermore, the ISIN of the enterprises will be crucial in filtering all other data. In other words, ISIN will serve as a unique identifier of each entity, and all other data on financial or carbon metrics will be matched in such a way that ISINs will be compared with the ISINs of the 702 firms from the emerging market region. As a result, solely the data on firms of interest will stay, while the firms without ISIN or from other regions will be removed.

The time-series data is clearly distinguished into two groups: monthly and annual. Data on market capitalization and total return index levels are reported at monthly frequencies, while the revenues and scope 1 and 2 emissions are annually reported. The data on market capitalization and total return indices are also available in annual form. Despite the fact that mainly annual data will be used for investment decisions, monthly data is still substantial for computing quantitative portfolio metrics such as returns, volatility, Sharpe ratio, and others.

Data cleaning is essential for ensuring economically meaningful and logical results. Firms without valid price observations are removed from the sample, while the enterprises with return indices below 0.5 are treated as missing values. If the value is missing between two observations, the prior metric is used, adhering to the forward fill principle. The missing values at the beginning of the sample are not modified and are usually due to firms not yet being listed or not yet reporting. If the observations are missing at the end of the sample, this typically suggests delisting of a firm, following the fact that the price is set to zero and implies the return of -100% recorded in the following month.

The total return index is used for computing monthly returns of enterprises. The simple returns are used instead of log returns due to the advantage of simple returns in constructing portfolios and calculating investment performance. In contrast, log returns are preferred in continuous compounding and stochastic calculus but not in portfolio management since they do not aggregate linearly across assets in a portfolio.

This is the simple return formula that was used:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1)$$

where R_t denotes the monthly return and P_t and P_{t-1} are the current and previous month's total return index levels, respectively. The total return index is a price index that includes dividend payments.

After computing the returns, to mitigate the potential artificially low portfolio volatility, firms that during at least 50% of monthly observations experience returns of zero, implying no price changes (due to forward fill), are removed from the investment universe. Therefore, this stale price filter prevents the effect of artificial unchanged prices from generating a biased volatility and, correspondingly, resulting in calculated weights in a portfolio based on a false assumption of low volatility.

3.2 Portfolio Construction

The investment set is a core of our analysis. An investment set is a list of companies at the end of the year in which one can invest. For the purpose of avoiding look-ahead bias, the portfolio is

constructed strictly out-of-sample, meaning that only past information up to the point we make investment decisions is used. As an example, at the end of the year 2013, only the previous data up to December of 2013 are utilized for the investment decisions in 2014. As our portfolio formation started in 2014, the period from the beginning of 2004 to the end of 2013 acted as our estimation window, which consisted of 120 months of observations. Based on these dates, the expected return vector and covariance matrix are computed. The criteria firms must respect in order to be in the investment set are, namely, belonging to the emerging markets region, a minimum of 3 years of monthly data on returns, and availability of carbon data for the needed years. As mentioned in section 3.1, firms that exhibit stale prices over the estimation window are excluded from the investment set.

The expected returns are computed following a historical approach by averaging 10 years of data up to year Y , where $\hat{\mu}_Y$ represents the expected return:

$$\hat{\mu}_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} R_{t-k} \quad (2)$$

The covariance matrix Σ_Y is obtained also by observing 10 years of data:

$$\Sigma_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} (R_{t-k} - \hat{\mu}_Y)' (R_{t-k} - \hat{\mu}_Y) \quad (3)$$

The covariance matrix is estimated with the Maximum Likelihood Estimator (MLE), usually used in finance, which divides by the sample size τ , in contrast with the statistical approach of dividing by $\tau - 1$.

The minimum variance portfolio, denoted as $P_{os}^{(mv)}$, is solved by optimizing the following function, imposing a condition on weights to be non-negative:

$$\begin{aligned} \min_{\alpha_Y} \quad & \sigma_{p,Y}^2 = \alpha_Y^\top \Sigma_Y \alpha_Y \\ \text{s.t.} \quad & \alpha_Y^\top e = 1, \quad e = (1, \dots, 1)^\top \\ & \alpha_{i,Y} \geq 0 \quad \text{for all } i \end{aligned} \quad (4)$$

where $\sigma_{p,Y}^2$ represents the portfolio variance (portfolio risk) in year Y , and $\alpha_{i,Y}$ represents the weight of the portfolio invested in firm i at the end of year Y .

The reason we only focus on non-negative weights, implying only long positions, stems from the fact that it is impossible to account for carbon emissions while holding a short position. In other words, it is a simplified assumption to make the interpretation of carbon footprint more feasible and intuitive. As previously stated, strong adherence to the out-of-sample principle is implemented while allocating assets at the end of each year for the subsequent year.

The dynamic nature of the following optimization problem is reflected in the rolling strategy, where the portfolio is reconstructed every year, starting from the end of 2013 to the end of 2024, which yields 12 iterations. Moreover, the optimization is conducted at the end of each year, but the performances are computed every month, resulting in updating weights on a monthly basis. In other words, a portfolio moves simultaneously with the market, which is best depicted in the drift formula:

$$\alpha_{i,t+k-1} = \alpha_{i,t+k-2} \times \frac{1 + R_{i,t+k-1}}{1 + R_{p,t+k-1}} \quad (5)$$

where $\alpha_{i,t+k-1}$ is the updated share of the portfolio's wealth invested in firm i after the realization of return in month $t + k - 1$, while $\alpha_{i,t+k-2}$ is the previous weight invested in firm i before the return realization. $R_{i,t+k-1}$ and $R_{p,t+k-1}$ denote the firm-level return and the portfolio return over month $t + k - 1$, respectively.

This generates 144 monthly ex-post portfolio returns, from the beginning of 2014 until the end of 2025, which will later be the basis for computing annualized performance metrics needed to calculate the Sharpe ratio, average return, maximum and minimum returns, and volatility.

In order to evaluate the performance of our minimum-variance portfolio, the Value-Weighted portfolio is constructed, serving as a benchmark. We denote it as $P_{oos}^{(vw)}$. A VW portfolio represents the outcome of passive investing, namely, what returns would one observe by simply holding the market? A VW portfolio is formed according to the market capitalization of each enterprise, where look-ahead bias is also mitigated by incorporating only past market capitalization metrics. To ensure consistency, the same firms that were in the investment set for the MV portfolio are used for generating the VW portfolio. Logically, only firms with positive market capitalization enter the investment set.

$$R_{t+1}^{(vw)} = \sum_{i=1}^N w_{i,t} R_{i,t+1} \quad (6)$$

where $R_{t+1}^{(vw)}$ denotes the return of the value-weighted portfolio in month $t + 1$, $R_{i,t+1}$ denotes the return of firm i in the same month, and $w_{i,t}$ denotes the share of firm i in the portfolio at the end of month t .

The mentioned weights are calculated by dividing the market capitalization of the company of interest by the total market capitalization of all firms in our investment set:

$$w_{i,t} = \frac{Cap_{i,t}}{\sum_{j=1}^N Cap_{j,t}} \quad (7)$$

3.3 Carbon Metrics

For the second part of our project, the climate-aware component is added. The aim of the second part is to reduce CO₂ emissions by 50% while impacting the financial performance as minimally as possible. Carbon intensity is a metric calculated for all companies in the investment set that captures the carbon emissions produced by the firm per unit of revenue. In our case, both scope 1 and scope 2 emissions are taken into consideration. Scope 1 emissions are direct emissions, while scope 2 emissions are top-tier indirect emissions, which arise from the consumption of purchased electricity and heat by a company. Since the carbon intensity is expressed in tonnes of CO₂ emissions per million of revenue, to obtain correct results, the data on revenues in thousands was divided by 1000. Furthermore, we computed weighted-average carbon intensity and the carbon footprint of the portfolio, which correspond to the annual amount of carbon emissions attributed to the investor per million dollars invested:

$$WACI_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CI_{i,Y} \quad (8)$$

where $CI_{i,Y}$ denotes the carbon intensity of firm i , and $WACI_Y^{(p)}$ refers to the weighted average carbon intensity of portfolio p in year Y .

Carbon footprint is computed as followed:

$$CF_Y^{(p)} = \frac{1}{V_Y} \sum_{i=1}^N o_{i,Y} E_{i,Y} \quad (9)$$

where V_Y represents the total value of the portfolio in a given year, $E_{i,Y}$ Scope 1 and Scope 2 emissions in the year of interest, and $o_{i,Y}$ denotes the ownership share of enterprise i that is held by the portfolio.

Moreover, the ownership share of the firm of interest is calculated by dividing the value invested in the firm, measured in dollars, by the firm's market capitalization in the year Y :

$$o_{i,Y} = \frac{V_{i,Y}}{Cap_{i,Y}} \quad (10)$$

$$V_{i,Y} = \alpha_{i,Y} V_Y \quad (11)$$

$$V_Y = \sum_{i=1}^N V_{i,Y} \quad (12)$$

Both metrics will be calculated for the minimum-variance portfolio and value-weighted portfolio, assuming the beginning wealth in 2013 amounts to 1 million dollars.

Furthermore, an optimal long-only portfolio will be generated, with a feature of a carbon footprint 50% lower than that of the minimum-variance portfolio. The novelty of this portfolio is the introduction of the carbon constraint, but the same objective of minimizing risk is maintained. Other characteristics of the portfolio are that all capital must be invested and no short selling is allowed. Portfolio weights are computed every year, considering the reduction of carbon footprint by solving this constrained optimization problem:

$$\begin{aligned} \min_{\alpha_Y} \quad & \sigma_{p,Y}^2 = \alpha_Y^\top \Sigma_Y \alpha_Y \\ \text{s.t.} \quad & CF_Y^{(p)} \leq 0.5 \times CF_Y^{(P_{oos}^{(mv)})} \\ & \alpha_Y^\top e = 1 \\ & \alpha_{i,Y} \geq 0 \quad \text{for all } i \end{aligned} \quad (13)$$

where $CF_Y^{(p)}$ indicates the carbon footprint of the constructed carbon-constrained portfolio, while $CF_Y^{(P_{oos}^{(mv)})}$ denotes the carbon footprint of the unconstrained minimum-variance portfolio. The derived minimum variance portfolio will be referred to as $P_{oos}^{(mv)}(0.5)$.

In contrast with this strategy of actively managing risk, the tracking error approach introduces the strategy of staying as close as possible to the benchmark portfolio while reducing carbon emissions. Tracking error measures the deviation of the portfolio from the benchmark for each year:

$$\begin{aligned}
\min_{\alpha_Y} \quad & TE_{p,Y} = \sqrt{(\alpha_Y - \alpha_Y^{(vw)})^\top \Sigma_Y (\alpha_Y - \alpha_Y^{(vw)})} \\
\text{s.t.} \quad & CF_Y^{(p)} \leq 0.5 \times CF_Y^{(P^{(vw)})} \\
& \alpha_Y^\top e = 1 \\
& \alpha_{i,Y} \geq 0 \quad \text{for all } i
\end{aligned} \tag{14}$$

where $TE_{p,Y}$ represents the tracking error of the portfolio in year Y , and $CF_Y^{(P^{(vw)})}$ depicts the carbon footprint of the value-weighted benchmark portfolio.

The carbon constraint is imposed, as in the previous strategy, where the carbon footprint is reduced in comparison with the minimum-variance portfolio. However, in this case, the reduction target is defined relative to the value-weighted benchmark portfolio rather than the minimum-variance portfolio. Accordingly, the constructed portfolio is labeled as $P_{oos}^{(vw)}(0.5)$.

Conversely, the final portfolio will address the carbon issue by gradually reducing the carbon footprint, rather than through a one-time adjustment. The net zero portfolio, denoted as $P_{oos}^{(vw)}(NZ)$, continuously reduces the carbon footprint by $\theta = 10\%$ each year, starting from Dec 2013 to Dec 2024. The passive investor perspective is retained, therefore resembling the $P_{oos}^{(vw)}(0.5)$, with the key distinction of replacing the fixed 50% carbon reduction with the continuously declining carbon budget:

$$CF_Y^{(p)} \leq (1 - \theta)^{Y - Y_0 + 1} \times CF_{Y_0}^{(P^{(vw)})} \quad \text{for } Y = 2013, \dots, 2024, Y_0 = 2013 \tag{15}$$

4 Results and Discussion

4.1 Part I

4.1.1 Minimum-Variance Portfolio

Table 1: Performance Comparison of MV and VW Portfolios

Metric	MV	VW
Annualized Average Return	7.69%	8.24%
Annualized Volatility	10.39%	15.54%
Sharpe Ratio	0.57	0.42
Minimum Monthly Return	-6.70%	-16.77%
Maximum Monthly Return	14.52%	13.41%

This table summarizes the key results for the Minimum Variance and Value-Weighted portfolios. It includes annualized returns, volatility, Sharpe ratios, and the maximum and minimum monthly returns observed over the sample period.

The first part of our analysis focuses on the out-of-sample performance of an MV portfolio that focuses solely on minimizing risk without carbon constraints. The portfolio's financial performance is measured during 144 months, namely, the period between the beginning of 2014 and the end of 2025. The MV portfolio exhibits an annual average return of 7.69% with a corresponding annual volatility of 10.4% throughout the observation window. Furthermore, the fluctuations

of the monthly returns span from 14.52%, as the highest monthly return observed, to -6.7%, presenting the lowest monthly return recorded. The Sharpe ratio equals 0.57, which illustrates a solid risk-adjusted performance, where for each additional unit of risk taken, investors experience 0.57 units of excess return.

The VW portfolio, on the other hand, is not subject to optimization but rather passively follows the market, and weights are determined by firms' relative market capitalizations. In our analysis, it behaves as the benchmark for assessing the performance of the MV portfolio. The VW portfolio, during the same timespan, achieves a slightly higher average annual return of 8.24%, but with the resulting greater annual volatility, which amounts to 15.54%. This tradeoff between returns and volatility is best reflected in the VW portfolio's Sharpe ratio, which stands at 0.42. Comparing with the MV portfolio, the VW portfolio experiences lower excess return for every unit of risk, despite the slightly higher average return. Logically, this is the consequence of substantially lower volatility of the MV portfolio compared with the benchmark portfolio. The oscillations are also of greater dispersion, especially on the negative side, where the lowest monthly return is recorded at -16.77%. The highest monthly returns amount to 13.41%, which is moderately lower compared with the MV portfolio. That arises from the fact that the VW portfolio passively follows the market and is, as such, more exposed to the cyclical nature of large firms, resulting in higher uncertainty.

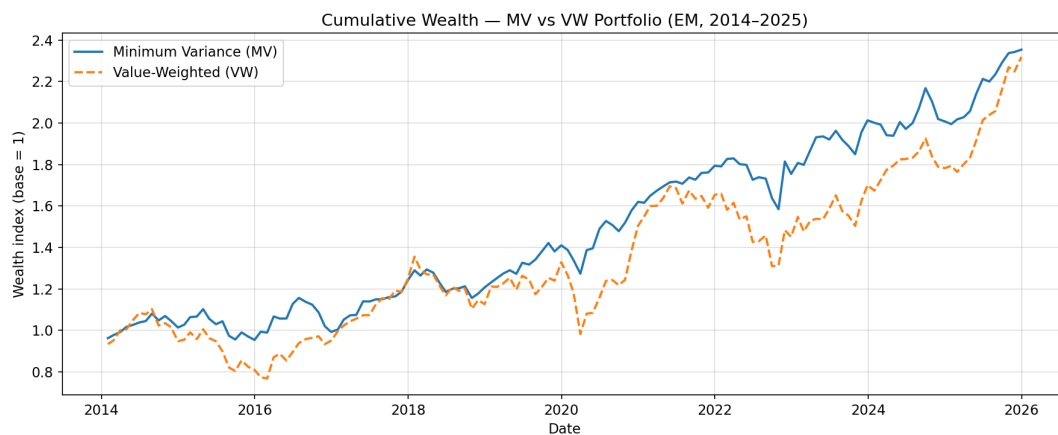


Figure 1: Cumulative returns of the minimum variance and value-weighted portfolios

This figure illustrates the cumulative wealth trajectory between 2014 and 2025 for the Minimum Variance and Value-Weighted portfolios. The initial wealth is normalized to 1.

As depicted in Figure 1, the cumulative wealth trajectory of the MV portfolio appears smoother throughout the sample period compared to the VW portfolio's performance, which exhibits greater fluctuations.

4.1.2 Carbon Characteristics of Baseline Portfolios

In order to grasp the benefits of carbon constraints, which will be examined in the following section, the insight on the carbon characteristics of the baseline MV and VW portfolios is needed. Weighted average carbon intensity and carbon footprint are key indicators that are utilized to convey the carbon exposure of the baseline portfolios.

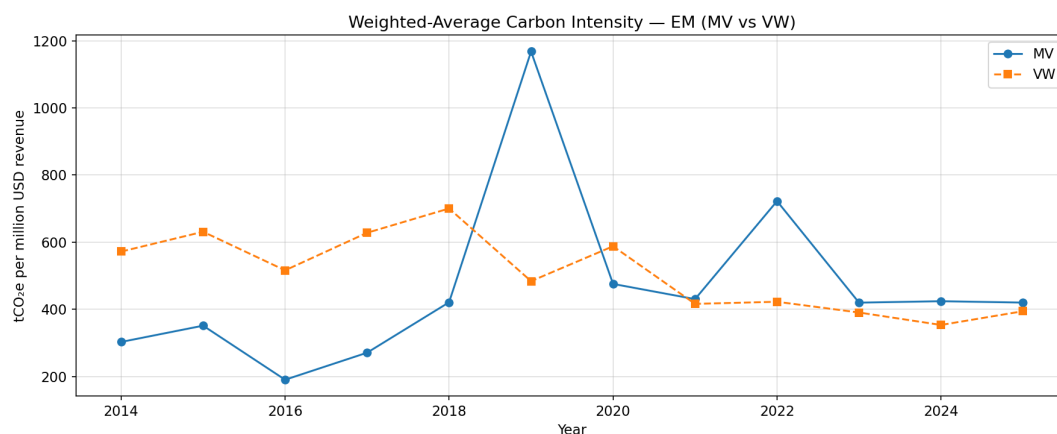


Figure 2: Comparison of WACI between the minimum variance and value-weighted portfolios

This figure illustrates the evolution of WACI for the Minimum Variance and Value-Weighted portfolios between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD of revenue.

Figure 2 depicts the evolution of weighted average carbon intensity during the sample period, comparing the two portfolios. The figure suggests that MV portfolios on average exhibit lower WACI, especially in the beginning period. Nonetheless, during 2019 and 2022, the MV portfolio experienced fluctuations of much larger magnitude, resulting in a higher WACI compared to the VW portfolio.

To facilitate the understanding of the carbon metrics, the following numbers are introduced. The MV portfolio exhibits on average a WACI of 466.42, indicating that companies included in the portfolio emit roughly 466.2 tons of CO₂ equivalent emissions per million USD of revenue. Conversely, the VW portfolio amounts on average to a WACI of 508.05, implying higher CO₂ equivalent emissions per million of revenue, for approximately 42 tonnes, compared with the MV portfolio.

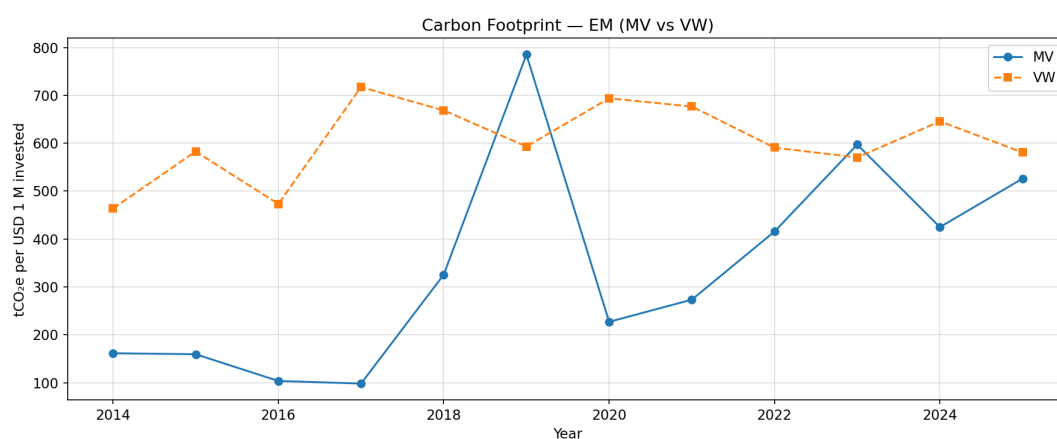


Figure 3: Comparison of carbon footprint between the minimum variance and value-weighted portfolios

This figure illustrates the evolution of the carbon footprint for the Minimum Variance and Value-Weighted portfolios between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD invested.

Figure 3 visualizes the progression of the alternative emission attribution measure—the carbon footprint—for both portfolios during the 11-year period. Similarly to the previous graph, the MV portfolio has, throughout the majority of the sample period, experienced a significantly

lower carbon footprint. Again, during 2019, the MV portfolio exhibited a higher level of carbon footprint, while a moderately higher value is also observed in 2023.

More precisely, the MV portfolio has, on average, a carbon footprint of 341.5, implying that approximately 341 tons of CO₂ equivalent emissions per 1 million USD invested are attributable to the investor. The VW portfolio, on the other hand, demonstrates a substantially higher average carbon footprint value of 604.69.

When it comes to top ten contributors of CO₂ emissions, measured by the WACI indicator, table 2 emphasizes the names of the companies and their corresponding weights in the baseline portfolios.

Table 2: Top 10 contributors to the portfolio's weighted average carbon intensity in 2024

ISIN	Company	Sector	Weight (%)	CI	WACI contrib.
PEP702101002	ENGIE ENERGIA PERU	Utilities	3.68	3415.7	125.80
PK0053401011	FAUJI FERTILIZER COMPANY	Materials/Chemicals	2.57	3458.0	88.77
PHY5764J1483	MANILA ELECTRIC	Utilities	3.29	1669.8	54.91
CNE1000002S8	COSCO SHIP.EN.TRSP. H	Transportation	3.15	1283.0	40.47
TW0002412004	CHUNGHWA TELECOM	Telecommunications	17.76	97.1	16.74
PEP700511004	ORYGEN PERU	Utilities	0.78	2032.3	15.82
TW0004904008	FAR EASTONE TELECOM.	Telecommunications	13.95	98.4	13.73
PHY751061151	SAN MIGUEL	Consumer Staples	0.89	958.8	8.57
INE059A01026	CIPLA	Pharmaceuticals	2.64	308.3	8.15
INE361B01024	DIVIS LABORATORIES	Pharmaceuticals	0.89	762.8	6.80

This table lists the top 10 contributors to the portfolio's WACI in 2024. It includes the companies, their industries, portfolio weights, carbon intensity (CI), and WACI contributions.

Table 2 highlights that the main contributors to the portfolio's WACI are firms operating in the energy, transportation, and industrial sectors, which are traditionally associated with high carbon emissions. Companies such as *ENGIE ENERGIA PERU*, *FAUJI FERTILIZER COMPANY*, *MANILA ELECTRIC*, and *COSCO SHIP.EN.TRSP.H* exhibit particularly high carbon intensity levels, as reflected by their CI values, which are substantially higher than those of the other firms in the list.

Despite these elevated carbon intensity levels, their contribution to portfolio-level WACI remains partially limited by their relatively lower portfolio weights. Conversely, companies such as *CHUNGHWA TELECOM* and *FAR EASTONE TELECOM*, although displaying relatively low carbon intensity levels, still contribute to WACI at levels close to those of high-emission companies due to their large portfolio weights.

From an economic perspective, these results suggest that carbon exposure in emerging markets remains strongly linked to the energy, transportation, and industrial sectors, which continue to play a central role in economic development and portfolio allocation.

To sum up, the minimum-variance portfolio shows lower carbon exposure than the benchmark portfolio, but the results also suggest that risk minimization alone is not enough to ensure the systematic long-term reduction of the carbon exposure.

4.2 Part II

4.2.1 Carbon-Constrained Portfolios

In this stage of the analysis, carbon constraints are introduced with the purpose of enhancing the portfolios' climate performance without significantly compromising their financial performance. The objective is to compare the ex-post performance of the minimum-variance and the tracking-error portfolio (relative to the VW benchmark), following the implementation of a 50% carbon footprint reduction.

Figure 4 compares the cumulative wealth of four portfolios, namely the MV and VW portfolios, together with their carbon-constrained variants. A finding that is particularly notable is the outperformance of the VW -50% CF portfolio, which is represented by the red dashed line, during the sample period. The outperformance is specifically apparent during the final two years of observations. Furthermore, the key finding is that carbon footprint reduction does not appear to clearly or materially decrease the financial performance of the portfolios. What's more, the carbon-constrained VW portfolio constructed through tracking error minimization is associated with stronger performance. Regarding the MV-50% CF portfolio, its performance stayed very close to the unconstrained MV portfolio. However, despite the empirical evidence that carbon footprint reduction does not impose an evident financial penalty, the findings should be interpreted with caution since the sample-specific effects may be present.

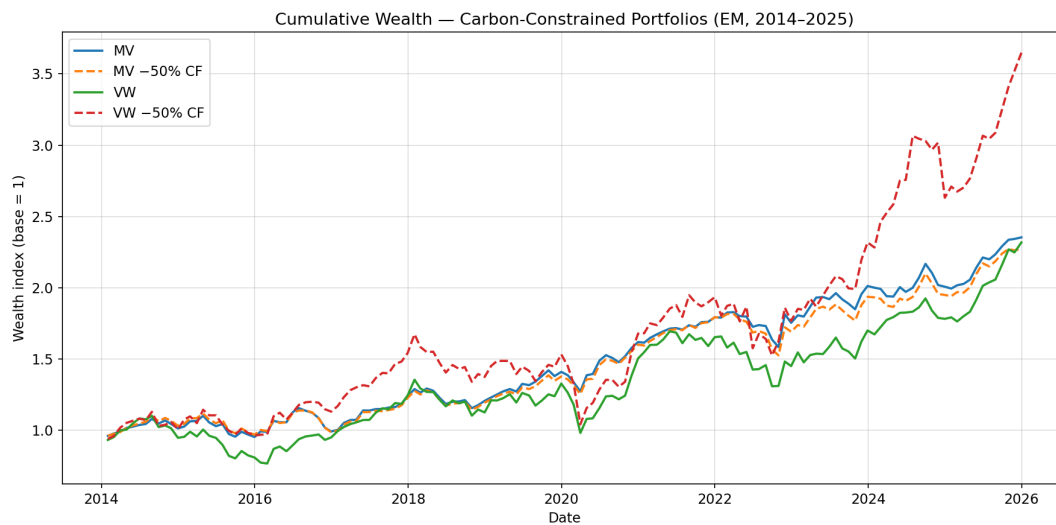


Figure 4: Cumulative performance of the carbon-constrained portfolios

This figure illustrates the cumulative wealth trajectory between 2014 and 2025 for the Minimum Variance and Value-Weighted portfolios, both with and without a 50% carbon footprint (CF) constraint. The initial wealth is normalized to 1.

Figures 5 and 6 illustrate the climate dimension of carbon-constrained portfolios. Figure 5 specifically presents the impact of the carbon footprint reduction on the portfolio's weighted average carbon intensity. An evident finding is that carbon footprint reduction in general results in lower WACI. The impact is not identical in magnitude to the reduction in carbon footprint, which is reasonable since the two are different carbon measures. The constrained minimum-variance portfolio exhibits lower WACI across the entire evaluation period in comparison with its unconstrained version. For the constrained value-weighted portfolio, lower WACI is generally observed, except in 2024. The underlying findings suggest that constructing 50% lower carbon footprint portfolios also tends to shift investments toward firms with lower carbon intensity.

Nonetheless, the reduction in WACI is not constant over time, since WACI and carbon footprint capture the different forms of carbon exposure.

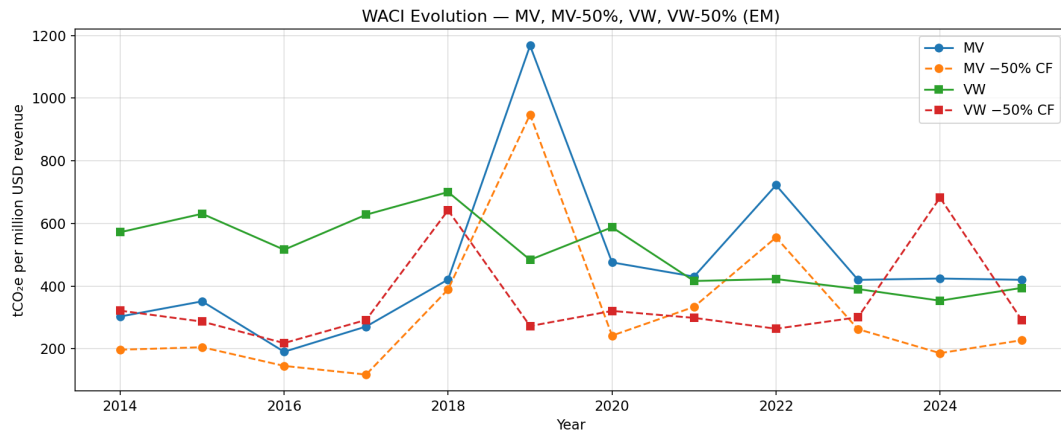


Figure 5: Weighted average carbon intensity of the carbon-constrained portfolios

This figure illustrates the evolution of WACI for the Minimum Variance and Value-Weighted portfolios, both with and without carbon constraints, between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD of revenue.

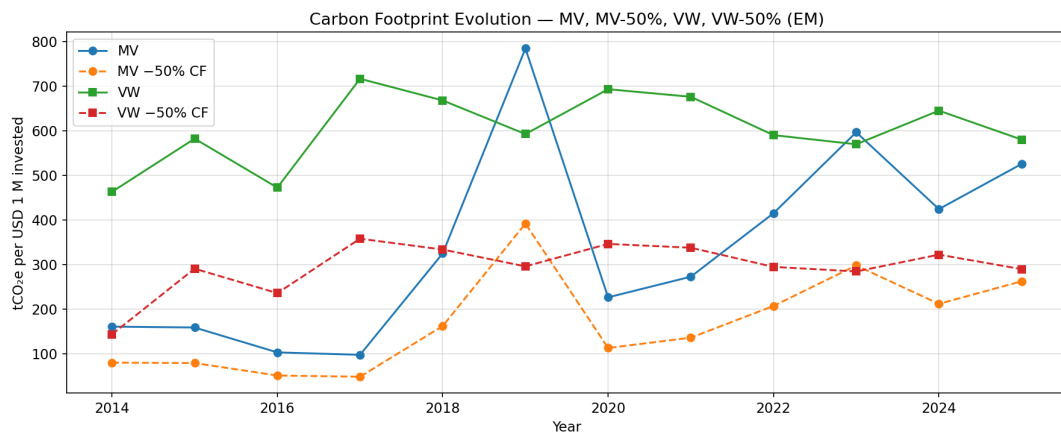


Figure 6: Carbon footprint of the carbon-constrained portfolios

This figure illustrates the evolution of the carbon footprint for the Minimum Variance and Value-Weighted portfolios, both with and without carbon constraints, between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD invested.

Figure 6 focuses on the implications for the carbon footprint itself. It is evident that the carbon footprint is indeed consistently lower in constrained portfolios, which underlines the reduction of investor-attributed emissions. Furthermore, the effectiveness of the 50% carbon footprint restriction is best reflected in its ability to substantially limit the carbon exposure during the more extreme periods, such as in 2019 for the MV portfolio. The constrained portfolios generally follow the movements of their unconstrained counterparts but at lower levels, suggesting the consistent downward impact of carbon restriction on the carbon footprint levels.

4.2.2 Net-Zero Portfolio

The net-zero portfolio approach shifts the nature of the carbon footprint constraint from being fixed to becoming gradual in nature. The idea is to test whether the 10% annual reduction in

the portfolio's carbon footprint is resulting in satisfying financial performance compared to the VW benchmark and its constrained variant.

Figure 7 emphasizes that the net zero portfolio consistently outperforms the VW benchmark in regard to cumulative wealth, while it has almost identical dynamics with the VW -50% CF portfolio, although with slightly poorer performance. The great discrepancy between the unconstrained VW portfolio and the net zero portfolio underlines that the net zero approach (similar to a 50% carbon footprint reduction) does not appear to impose a clear negative repercussion on financial performance.

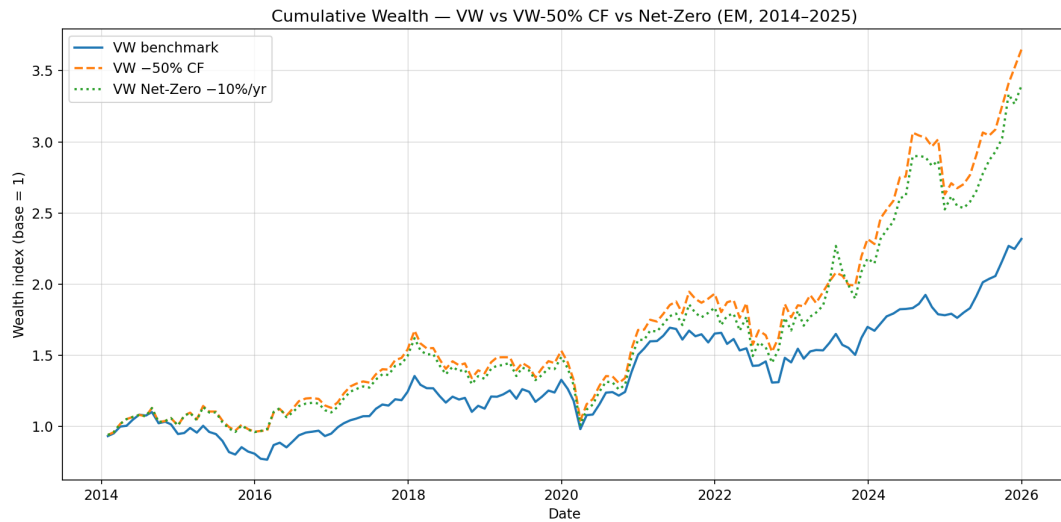


Figure 7: Cumulative performance of the net-zero portfolio

This figure illustrates the cumulative wealth trajectory between 2014 and 2025 for the Value-Weighted benchmark portfolio, the 50% carbon footprint constrained portfolio, and the Net-Zero portfolio. The initial wealth is normalized to 1.

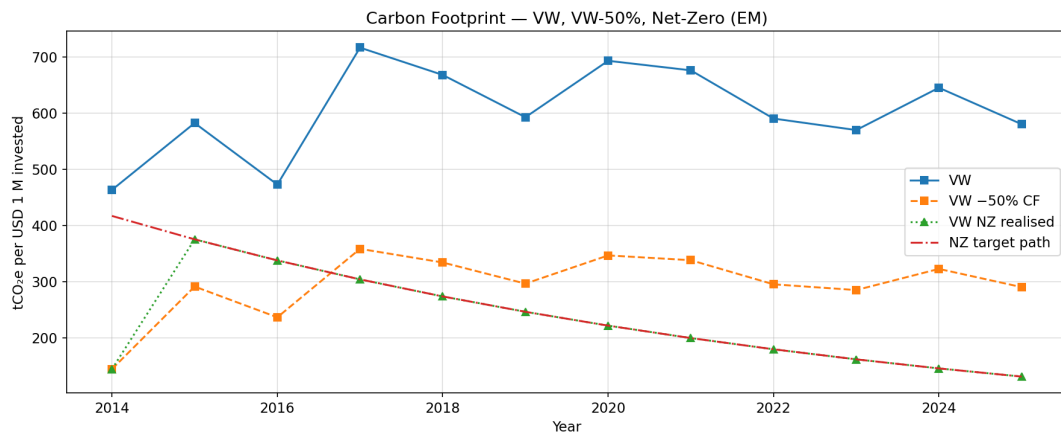


Figure 8: Carbon footprint trajectory of the net-zero portfolio

This figure illustrates the evolution of the carbon footprint for the Value-Weighted portfolios, both with and without carbon constraints, together with the Net-Zero target path, between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD invested.

The development of the carbon footprint is illustrated in figure 8, where for each of the three portfolios, the levels of carbon footprint over the observed years are depicted. Most importantly, the carbon footprint is reduced every year and follows the decarbonization target path, indicating a successful implementation of the net-zero objective. As shown in figure 8, the constant and

consistent reduction of the carbon footprint of the net zero portfolio starts in 2015. This is due to the set constraint that requires the carbon footprint to be lower than or equal to the target path, which acts as the upper bound rather than the target. In 2015, as the realized carbon footprint reaches the target path, the decreasing target path starts to drive the carbon footprint downward in subsequent years. Moreover, it may be inferred that the net-zero portfolio results in lower carbon footprint values in the majority of years, relative to the VW -50% CF portfolio, except for the beginning of the sample. That can be due to the persistent and compounding nature of the net-zero portfolio constraint, where the reduction is accumulated constantly and, in that way, outperforms in the long term the VW-constrained portfolio in the decarbonization process.

Figure 9 shows the trajectory of the weighted average carbon intensity, depending on the underlying portfolio. The figure suggests that the net-zero portfolio exhibits dominantly lower levels of WACI in contrast to the VW benchmark and, from 2019 onwards, in regard to the constrained VW portfolio. Logically, the impact of implementing a net-zero policy is not the same for the WACI as for the carbon footprint, which can be identified in the nonmonotonic dynamics of the WACI under the net-zero approach, contrary to the carbon footprint decreasing monotonically.

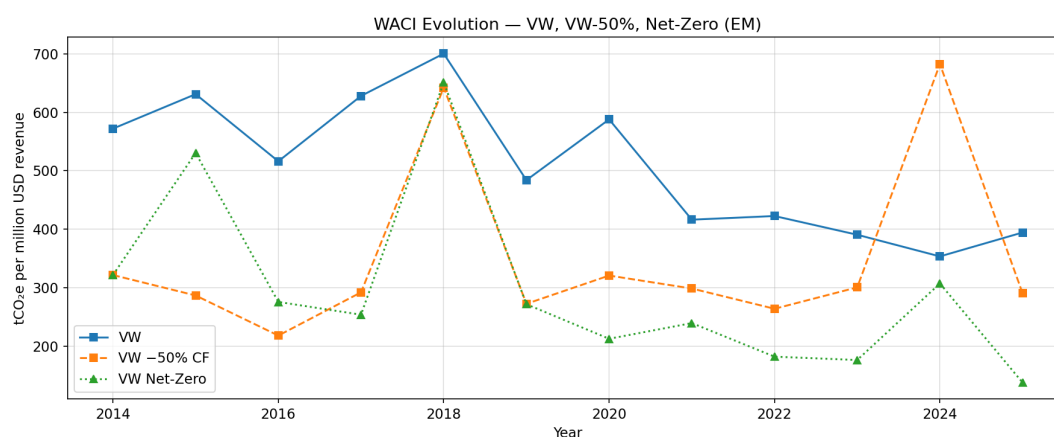


Figure 9: Weighted average carbon intensity (WACI) of the net-zero portfolio

This figure illustrates the evolution of WACI for the Value-Weighted portfolios, both with and without carbon constraints, between 2014 and 2025, measured in tons of CO₂ equivalent emissions per million USD of revenue.

However, the potential drawback can be the lack of diversification due to concentrating investments mostly in cleaner firms. Furthermore, the marginally lower financial performance of the net-zero portfolio compared to the constrained VW portfolio can be seen as a drawback, not giving enough incentive to implement the stricter rule without expected financial compensation.

Overall, according to the findings, the net-zero strategy represents a plausible carbon-tightening approach without a clear adverse influence on the portfolio's financial performance. The net zero portfolio beats the VW benchmark over most of the sample period while slightly underperforming relative to the VW 50% CF portfolio in financial terms. Together with its effectiveness in carbon reduction, a potential trade-off between climate and financial objectives can be identified. That is to say, the net-zero approach achieves stronger long-term carbon footprint reduction, but this stricter decarbonization path may involve an opportunity cost compared to the fixed 50% reduction strategy.

4.3 Comparative Analysis

Table 3: Performance Comparison of Carbon-Constrained Portfolios

Metric	MV	MV 50% CF	VW	VW 50% CF	VW Net-Zero
Annualized Average Return	7.69%	7.38%	8.24%	12.51%	12.04%
Annualized Volatility	10.39%	10.14%	15.54%	18.14%	20.18%
Sharpe Ratio	0.57	0.56	0.42	0.59	0.54
Minimum Monthly Return	-6.70%	-6.90%	-16.77%	-21.42%	-21.63%
Maximum Monthly Return	14.52%	12.91%	13.41%	15.38%	15.05%

This table summarizes the key results for the Minimum Variance and Value-Weighted portfolios, both with and without carbon constraints. It includes annualized returns, volatility, Sharpe ratios, and the maximum and minimum monthly returns observed over the sample period.

So far, we have examined several investment strategies, including MVP and VW portfolios, both with and without carbon constraints. The purpose of this section is to summarize all the results and draw conclusions regarding the implementation of carbon constraints in portfolio construction.

Table 3 summarizes the results for each strategy, taking into account returns, volatility, the Sharpe ratio, and the maximum and minimum one-month returns.

In terms of performance, the results are heterogeneous. We begin by examining strategies without carbon constraints. As mentioned in Section 4.1.1, the VW portfolio outperforms the MVP in terms of average returns (8.24% versus 7.69%), this is largely due to the minimization of volatility in the MVP. Regarding performance under carbon constraints, we observe superior results for the benchmark-constrained portfolios. Indeed, adding constraints does not necessarily imply a loss of performance, on the contrary, it leads to higher returns which is significantly higher than that of the MVP and the VW portfolio, with returns ranging from 12% to 12.5%. The MVP, on the other hand, underperforms the other strategies when subject to the constraint. Nevertheless, we must remain cautious with these results. Indeed, several factors may have contributed to these results, such as the investment period (2014 to 2025) or structural market factors that may have favored these results (COVID-19 or AI growth).

The carbon constraints applied to the benchmark result in outperformance. However, this performance comes at a cost, as evidenced by significantly higher volatility. While the MVP exhibits volatility levels of 10.39% and 10.14% under constraints, the VW portfolios with 50% CF and Net-Zero constraints exhibit volatility levels of 18.14% and 20.18%, respectively. This demonstrates that returns do not come without a cost and expose us to greater risks. In fact, we observe much more severe negative monthly returns under constraints, such as -21.42% for the 50% CF VW and -21.63% for the Net-Zero, compared to -6.70% for the MVP. This difference in risk between the portfolios may stem from their construction, indeed, the MVP optimizes volatility to be as low as possible, regardless of the constraint. As for the VW, it remains tied to the market weighting of companies and to an investment set constrained by CF constraints, which can affect diversification by reducing or excluding certain polluting sectors, and consequently increasing the exposure to other sectors.

Analyzing returns and volatility brings us to the central issue: the Sharpe ratio. If we had to choose one of these strategies, we would likely rule out the VW strategy, which has a Sharpe ratio of only 0.42. The others, however, remain relevant, indeed, the risk control provided by the MVP, both with and without constraints, results in strategies with ratios of 0.57 and 0.56. The VW strategies under constraints, meanwhile, reach 0.59 for the 50% CF and 0.54 for the Net Zero. The 50% CF VW strategy thus appears to be the most efficient risk-adjusted strategy,

however, it does not significantly outperform the MVP in terms of Sharpe ratio.

In short, decarbonization appears to be achievable and compatible with financial performance, but, it is important to note that the risk is significantly higher compared to an unconstrained portfolio, primarily due to the narrowing of the investment universe caused by the carbon screening constraints.

5 Limitations

The first limitation relates to the carbon data, regarding which we should note several points. First, we only consider Scope 1 and 2 emissions. Scope 3 emissions are not taken into account, even though they account for the majority of emissions in certain sectors (e.g., banking or insurance). Our portfolios therefore do not capture overall carbon exposure.

We also note that carbon data is very limited prior to 2010, as evidenced by our investment universes, which have grown progressively over the years. In fact, we started with just 245 companies in 2013, reaching 567 by 2024. Data quality must also be taken into account, as the way ESG reporting is conducted can vary from one firm to another.

Table 4: Number of Companies in the Investment Universe

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Companies	245	267	297	334	381	416	463	504	542	570	580	567

This table shows the evolution of the size of the investment universe over the sample period.

Another limitation of this project is the failure to account for transaction costs. These costs were not taken into consideration. However, they are of fundamental importance in asset management, particularly when rebalancing portfolios that experience high turnover. As such, these costs can significantly impact a portfolio's overall performance and skew the initial interpretation made without accounting for them.

Another limitation of this project is the estimation of expected returns. In this analysis, we based our calculations on expected returns derived from historical average returns. A more specific approach, such as using the CAPM or the multi-factor Fama-French model (with 3, 5, or 8 factors), would have yielded a different estimate, taking into account the risk factors that contribute to the expected return.

Furthermore, the results of this analysis may have been significantly influenced by the period under review (2014 to 2025). Indeed, the market has experienced several crises, such as the COVID-19 pandemic, which have at times altered the risk profile of certain assets and prompted strategic reallocations by investors. We also note the remarkable growth of AI in recent years, which has led to a potential stock market bubble, as well as the trade tensions waged by certain countries, such as China and the United States, which has significantly impacted stock market indices. All these events directly impact returns under carbon constraints and may partially skew the market conditions in our sample, making this analysis specific to this market period and difficult to generalize.

Finally, the last limitation concerns the region under study. This project focuses exclusively on emerging markets, where carbon dynamics may differ significantly from those in industrialized regions such as Europe or North America. Therefore, the results obtained in this analysis do not necessarily apply to other regions of the world. A specific study of other markets would provide

further insight into the effectiveness of incorporating carbon constraints into portfolios.

6 Conclusion

This report assessed whether the inclusion of decarbonization constraints based on Scope 1 and Scope 2 emissions adversely affects the financial performance of standard portfolio strategies. This analysis focused on firms in emerging markets and how the reduction of carbon emissions, measured through WACI and carbon footprint, influences financial outcomes. Namely, minimum-variance and value-weighted portfolios serve as unconstrained baseline portfolios, where the value-weighted portfolio represents the benchmark case. Firstly, a 50% carbon footprint reduction was implemented on the two baseline portfolios, resulting in two carbon-constrained portfolios. Secondly, the net-zero portfolio was introduced, where the carbon footprint is progressively reduced throughout the sample period. When it comes to unconstrained portfolios, the MV portfolio exhibited a higher Sharpe ratio and lower volatility, which is consistent with its objective of minimizing portfolio risk, while the VW portfolio experienced better average returns. Moreover, the MV displayed lower values of WACI and CF compared to the VW benchmark, but the sole risk minimization is not enough in itself to systematically implement decarbonization. For the constrained portfolios, in our analysis, carbon footprint restrictions reduce investor-attributed emissions and generally lower WACI in both MV and VW portfolios simultaneously without clearly worsening the financial performance. Interestingly, the VW -50% CF realized the highest Sharpe ratio and average return of all constructed portfolios, with the drawback of the high volatility, which is translated into the widest observed range of monthly returns. The net-zero portfolio successfully reduces the carbon footprint by 10% each year, although it comes with the highest volatility among all portfolios.

Therefore, the possibility of enhancing the climate profile of the portfolio is visible, although this may result in higher volatility due to concentration risk and sector biases. It is important to interpret these findings with caution, since the results are based only on the 2014-2025 period and exclusively derived from the analysis oriented towards companies operating in emerging markets. Moreover, the incorporation of transaction costs and Scope 3 emissions in the assessment could improve the robustness of the findings.

7 Use of Large Language Models (LLMs)

This project was carried out with supervised and limited use of artificial intelligence, in accordance with current university regulations.

The following tools were used for this project:

- ChatGPT (OpenAI, version 5.2)
- Copilot (Microsoft)
- DeepL
- Claude IA

Artificial intelligence has been used for the following aids:

- Learning and improving Python and L^AT_EX code.
- As a translation aid for certain words or phrases.
- To improve grammar and writing style.
- For debugging (particularly Python syntax and the use of other tools such as GitHub, Visual Studio Code, and Overleaf).
- Brainstorming and clarifying ideas.
- Searching for relevant academic sources related to research questions.

The content of the report and the code are the result of personal reflection, which encompasses all of its elements. The author assumes full responsibility for the work submitted.

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